INCH-POUND

MIL-PRF-8387OB (USAF)
31 December 1998
SUPERSEDING
MIL-F-8387OA (USAF)
22 September 1978

PERFORMANCE SPECIFICATION

FILTER ELEMENT, DISPOSABLE, 20 MICRON ABSOLUTE,

CONSTANT SPEED DRIVE, AIRCRAFT

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers constant speed drive (CSD) disposable filter elements that retain all particles larger than 20 microns. The elements are intended for use with lubricating oil conforming to MIL-L-7808 or MIL-L-23699 in the CSI)s specified in 6. 1.

1.2 Classification. CSD filter element part numbers (PN) should be as shown in table I and as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, 3001 Staff Drive, Suite IAEI-101A, Tinker AFB, OK 73145-3036, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1650

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TABLE I. Classification

Element part number	Figure
M83870-1	I
M83870-2	I
M83870-3	2
M83870-4	3
M83870-5	4
M83870-6	5
M83870-7	6
M83870-8	7
M83870-9	8
M83870-10	9

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are cited in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of these lists, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification. The following specification forms a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-L-7808 Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-L-23699 Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0- 156

(Unless otherwise indicated, copies of the above specification are available from the Defense Automated Printing Service, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

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2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE ARP 901	-	Bubble - Point Test Method
SAE AS 4716	-	Aerospace Standard, Gland Design, O-Ring and Other Elastomeric Seals (DoD-adopted)

(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4572	-	Hydraulic Fluid Power - Filters - Multi-Pass Method for Evaluating Filtration Performance

(Application for copies should be addressed to the International Organization for Standardization, Case postale 56, CH-1211 Geneve 20, Switzerland).

AMERICAN NATIONAL STANDARDS INSTITUTE/AMERICAN SOCIETY FOR QUALITY (ANSI/ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes (DoD-adopted)

(Application for copies should be addressed to American Society for Quality, P.O. Box 3066, Milwaukee, WI 53201-3066, or to the American National Standards Institute, 11 West 42nd Street, 13th Floor, New York, NY 10036.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The filter elements furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award (see 4.2 and 6.3).

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3.2 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle cost.

3.3 Materials. All materials shall be corrosion resistant or suitably treated to resist corrosion due to electrolytic decomposition, high temperature lubricating oil, and any other condition that may be encountered during operational use or storage. The use of toxic chemicals, hazardous substances, or ozone depleting chemicals shall be avoided, whenever feasible.

3.3.1 Compatibility. All materials shall be compatible with MIL-L-7808 or MIL-L-23699 lubricating oil. No copper shall be used.

3.4 Interface.

3.4.1 Dimensions. The filter element shall conform to the dimensions for the appropriate part number as designated by figures 1 through 9.

3.4.2 Type. The filter element shall be of the full-flow type. The flow through the filter element shall be from the outside of the filter to the inside.

3.5 Performance.

3.5.1 General. Filter element performance requirements are specified in table II.

3.5.2 Filter media. Filter media shall withstand thermal and pressure-flow cycling. In addition, there shall be no filter element media migration during operation.

3.5.3 Bubble point. The initial bubble point of the filter element shall be determined by testing, in accordance with the procedures in SAE ARP 901.

3.5.4 Fluid contamination. Filter material shall not be affected by or contaminate MIL-L-7808 or MIL-L-23699 lubricating oil.

3.5.5 Cold start. The filter element shall withstand 200 pounds per square inch (psi) differential pressure across the element for 10 flow cycles at -65° ± 5°F without permanent deformation or a bubble point decrease of greater than 10% of the initial bubble point.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

3.7 Item identification. The filter shall be permanently and legibly marked with the following information:

a. Manufacturer's name
b. Manufacturer's CAGE code

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c. Manufacturer's part number
d. Manufacturer's serial number
e. M83870-XX
f. Collapse pressure

TABLE II. Performance requirements

	M83870-1	M83870-2	M83870-3	M83870-4	M83870-5
Operating fluid temperature range (°F)	-65 to 350	-65 to 350	-65 to 350	-65 to 350	-65 to 350
Absolute rating (micron)	20	20	20	20	20
Rated flow (gal/min)	6.0	6.0	10.0	10.0	10.0
Efficiency					
Add quantity: F-9 beads (g)	0.30	0.30	0.30	0.30	0.30
Contaminant removal by weight (percent)	94	94	94	94	94
Pressure Buildup and Collapse					
Pressure drop (clean element) (psi)	10.5	10.5	10.5	10.5	10.5
AC dust add increments (g)	0.30	0.30	0.30	0.30	0.30
AC dust minimum weight (g)	2.75	6.30	3.60	3.00	3.00
Element pressure drop (psi)	40	40	40	40	40
Min collapse pressure (psi)	200	200	200	200	200
Flow Fatigue					
Hydraulic fluid temperature (°F)	350 ± 5	350 ± 5	350 ± 5	350 ± 5	350 ± 5
Flow rate (gal/min)	6.0	6.0	10.0	10.0	10.0
Cycling rate (cpm)	300	300	300	300	300
(a) Number of fatigue cycles Element pressure drop (psi)	1,125,000 20	1,125,000 20	1,125,000 20	1,125,000 20	1,125,000 20
(b) Number of fatigue cycles Element pressure drop (psi)	112,000 50	112,000 50	112,000 50	112,000 50	112,000 50
(c) Number of fatigue cycles Element pressure drop (psi)	12,000 100	12,000 100	12,000 100	12,000 100	12,000 100
(d) Number of fatigue cycles Element pressure drop (psi)	500 200	500 200	500 200	500 200	500 200
Max Particle Passed (micron)	20	20	20	20	20
Add rate F-9 beads (g)	0.30	0.30	0.30	0.30	0.30
Note: AC dust is standardized fine air cleaner dust.					

TABLE II. Performance requirements (cont'd)

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	M83870-6	M83870-7	M83870-8	M83870-9	M83870-10
Operating fluid temperature range (°F)	-65 to 350	-65 to 350	-65 to 350	-65 to 350	-65 to 350
Absolute rating (micron)	20	20	20	20	20
Rated flow (gal/min)	10.0	3.0	10.0	6.0	7.5
Efficiency					
Add quantity: F-9 beads (g)	0.30	0.30	0.30	0.30	0.30
Contaminant removal by weight (percent)	94	94	94	94	94
Pressure Buildup and Collapse					
Pressure drop (clean element) (psi)	10.5	10.5	10.5	10.5	10.5
AC dust add increments (g)	0.30	0.30	0.30	0.30	0.30
AC dust minimum weight (g)	3.00	1.10	2.40	1.75	2.40
Element pressure drop (psi)	40	40	40	40	40
Min collapse pressure (psi)	200	200	200	200	200
Flow Fatigue					
Hydraulic fluid temperature (°F)	350 ± 5	350 ± 5	350 ± 5	350 ± 5	350 ± 5
Flow rate (gal/min)	10.0	3.0	10.0	6.0	7.5
Cycling rate (cpm)	300	300	300	300	300
(a) Number of fatigue cycles Element pressure drop (psi)	1,125,000 20	1,125,000 20	1,125,000 20	1,125,000 20	1,125,000 20
(b) Number of fatigue cycles Element pressure drop (psi)	112,000 50	112,000 50	112,000 50	112,000 50	112,000 50
(c) Number of fatigue cycles Element pressure drop (psi)	12,000 100	12,000 100	12,000 100	12,000 100	12,000 100
(d) Number of fatigue cycles Element pressure drop (psi)	500 200	500 200	500 200	500 200	500 200
Max Particle Passed (micron)	20	20	20	20	20
Add rate F-9 beads (g)	0.30	0.30	0.30	0.30	0.30
Note: AC dust is standardized fine air cleaner dust.					

3.8 Filter element pressure relief valve.

3.8.1 M83870-6 pressure relief valve (see figure 5). The cracking (opening) pressure of the pressure relief valve shall be 100 ± 10 psi. The differential pressure across the pressure relief valve shall be a maximum of 160 psi when the test fluid flow rate through the pressure relief valve is 11 gal/min. The pressure relief valve shall reseat (close) at a minimum pressure of 65 psi.

3.8.2 M83870-7 pressure relief valve (see figure 6). The cracking (opening) pressure of the pressure relief valve shall be 50 ± 10 psi. The differential pressure across the pressure relief valve shall be a maximum of 75 psi when the test fluid flow rate through the pressure relief valve is 3 gal/min. The pressure relief valve shall reseat (close) at a minimum pressure of 38 psi.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows (see 6.2).

a. Qualification inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 Qualification inspection. If required (see 3.1 and 6.3), qualification inspection shall be performed on four filter elements of each part number. The filter elements shall be subjected to the tests indicated in table III.

TABLE III. Qualification tests

Filter element no. 1 (lowest initial bubble point)	Filter element no. 2 (highest initial bubble point)	Filter element no. 3 (2nd lowest initial bubble point)	Filter element no. 4	Test
Examination of product	Examination of product	Examination of product	Examination of product	4.6.1
Initial bubble point	Initial bubble point	Initial bubble point	Initial bubble point	4.6.2.1
Immersion	Immersion	Immersion	Immersion	4.4.3
Initial bubble point	Initial bubble point	Initial bubble point	Initial bubble point	4.6.2.1
Max particle passed	N/A	N/A	N/A	4.6.5
Cold start	N/A	N/A	N/A	4.6.3
Initial bubble point	N/A	N/A	N/A	4.6.2.1
N/A	Pressure buildup and collapse	N/A	N/A	4.6.6
N/A	Initial bubble point	N/A	N/A	4.6.2.1
N/A	N/A	Efficiency	N/A	4.6.4
N/A	N/A	Initial bubble point	N/A	4.6.2.1
N/A	N/A	Flow fatigue		4.6.7
N/A	N/A	Initial bubble point	N/A	4.6.2.1
N/A	N/A	N/A	Media and abrasion migration	4.6.8
N/A	N/A	N/A	Initial bubble point	4.6.2.1

4.3 Conformance inspection. Conformance inspection shall consist of the individual tests in 4.3.1 and the sampling tests in 4.3.2.

4.3.1 Individual. Each filter element shall be subjected to the examination in 4.6.1, except for dimensional analysis, the conformance bubble point test in 4.6.2.2, and, when applicable, the pressure relief valve test in 4.6.9.

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4.3.2 Sampling. A sample of filter elements shall be selected in accordance with ANSI/ASQ Z1.4 at an S-1 inspection level unless modified in the procurement document (see 6.2), and shall be subjected to the pressure build-up and collapse test in 4.6.6, and shall not exceed the pressure drop (clean element) value specified in table II.

4.4 Test conditions and preparation.

4.4.1 Test fluid. Unless otherwise specified, the test fluid used for all tests shall conform to MIL-L-7808 lubricating oil.

4.4.1.1 Test fluid cleanliness. Unless otherwise specified, the test fluid shall be pre-cleaned by circulation through a 5-micron absolute filter (see 6.4). Contamination in the test fluid shall be less than 0.0004g/2000 mL of test fluid.

4.4.1.2 Test fluid temperature. Unless otherwise specified, the test fluid temperature shall be 100° ± 10°F.

4.4.2 Temperature. Unless otherwise specified, the tests shall be conducted at room ambient temperature.

4.4.3 Immersion. After performing the initial bubble point test and prior to performing the qualification tests (see table III), filter elements shall be immersed in 350°F test fluid for 100 hours. All internal parts shall be in contact with the test fluid during this period. After immersion, the filter elements shall remain in the test fluid at room temperature until further tests are conducted.

4.4.4 Test air. The air used to pressurize a test setup shall be filtered through a membrane filter (0.8 micron absolute)(see 6.5).

4.4.5 Flow cycle. A flow cycle consists of increasing the differential pressure across the filter element from 0 to 200 psi and then decreasing to 0 psi.

4.4.6 Fatigue cycle. A fatigue cycle consists of increasing the differential pressure across the filter element (loaded with AC dust) from 0 to the element pressure drop in the flow fatigue part of table II and decreasing to 0 psi.

4.5 Requirements cross-reference matrix. Table IV provides a cross-reference matrix of the section 3 requirements tested or verified in the paragraphs below.

TABLE IV. Requirements cross-reference matrix

Requirement	Verification	Requirement	Verification
3.1	4.2	3.5.3	4.6.2
3.3	4.6.1, 4.6.7	3.5.4	4.6.4
3.3.1	4.6	3.5.5	4.6.3
3.4.1	4.6.1	3.6	4.6.1

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3.4.2	4.6.1	3.7	4.6.1
3.5.1	4.6	3.8.1	4.6.9.1
3.5.2	4.6.8	3.8.2	4.6.9.2

4.6 Tests.

4.6.1 Examination. The filter element shall be examined for compliance with requirements for materials, dimensions, and markings.

4.6.2 Bubble point.

4.6.2.1 Initial bubble point. The initial bubble point of each filter element undergoing qualification testing shall be determined in accordance with the procedures in SAE ARP 901. The lowest initial bubble point of the four qualification test filter elements is the value to be used in all subsequent tests referring to initial bubble point.

4.6.2.2 Conformance bubble point. Each filter element shall be subjected to the bubble point test in SAE ARP 901 and shall have a bubble point equal to or higher than the lowest initial bubble point established in 4.6.2.1.

4.6.3 Cold start. The filter element shall be installed in a non-bypassed test filter housing filled with test fluid, placed in a test system, and subjected to a temperature of –65° ± 5°F for four hours (a typical test system is shown in figure 10). At or below this test fluid temperature, 10 flow cycles shall be conducted with each cycle 15 ± 5 seconds in duration. The flow rate during this test shall not exceed 150% of rated flow (see table II) or be less than 0.5 gal/min. There shall be no filter media damage as evidenced by satisfactory completion of the initial bubble point (4.6.2.1), maximum particle passed (4.6.5), pressure buildup and collapse (4.6.6), and flow fatigue (4.6.7) tests.

4.6.4 Efficiency. The efficiency for filter elements is computed as specified in 4.6.4.1 and shall meet the contaminant removal by weight specified in the efficiency test in table II.

4.6.4.1 Efficiency computation.

$$\text{Efficiency} = \frac{A - (B - C)}{A} \times 100 \qquad (1)$$

where: A = add value
B = contaminant value
C = blank value

4.6.4.2 Contaminant value (B). The filter element shall be installed in a test filter housing and placed in a test system which has been previously flushed with pre-cleaned test fluid (see 4.4.1.1) (a typical test system is shown in figure 11). Test fluid (2,000 mL) shall be added to the test system. Then, a slurry containing the add value of contaminants shown in table II shall be mixed with the test fluid. The test fluid, at the rated flow specified in table II, shall be passed through the filter element and then through a previously cleaned and weighed 0.8 micron absolute filter

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membrane (see 6.6). The test fluid shall be collected and the filter membrane weighed. The difference in weight between the filter membrane before and after this procedure is the weight of contaminant passing through the filter element and is the contaminant value (B) in equation 1.

4.6.4.3 Add value (A). A free-flow dummy element shall be installed in a test filter housing and the procedure in 4.6.4.2 shall be repeated four times. The add value (A) in equation 1 is the average of the weights of contaminant collected in the four runs. The weight of the contaminant collected in each of the four runs shall be 95% or more of the weight of the contaminant added.

4.6.4.4 Blank value (C). The filter element shall be installed in a test filter housing and, with no contaminant added, the procedure in 4.6.4.2 shall be repeated. The weight of contaminant collected is the blank value (C) in equation 1. The blank value is the weight of contaminant attributed to the test system and filter element and shall be less than 0.0007 gram.

4.6.4.5 Multi-pass efficiency testing. If required, multi-pass efficiency testing, in accordance with ISO 4572, will be performed (see 6.2 and 6.3.1).

4.6.5 Maximum particle passed. The maximum particle passed is determined by following the procedure in 4.6.4.2 using the add value of contaminant specified in table II. The element with the lowest initial bubble point shall be used in this test. The largest particle size passed shall be no greater than the value specified in table II.

4.6.6 Pressure buildup and collapse. The filter element with the highest initial bubble point shall be installed in a test filter housing and placed in a test system filled with test fluid pre-cleaned in accordance with 4.4.1.1 (a typical test system is shown in figure 12). The rated flow shown in table II shall be maintained during this test. The clean-up filter shall be bypassed during this test. The pressure drop across the clean element shall be measured to verify that the clean element pressure drop does not exceed the value in table II. Add AC dust, in slurry, Through the dust valve in the incremental amounts specified in table II at 4 minute intervals. When the minimum AC dust in table II is added, the element pressure drop in table II shall not be exceeded. Additional AC dust shall be added until the minimum collapse pressure in table II is obtained. There shall be no sign of structural failure indicated through a visual inspection and the bubble point test in 4.6.2.1.

4.6.7 Flow fatigue. The filter element shall be installed in a non-bypass type test filter housing, placed in a test system and subjected to fatigue cycles with the test fluid at table II temperature and flow rate. The cycling rate and the number of fatigue cycles at each differential pressure are specified in table II.
There shall be no evidence of damage as indicated by performing the bubble point test in 4.6.2.1 (to verify that the bubble point has decreased by no more than 10%) and by satisfactory completion of the media migration test in 4.6.8.

4.6.8 Media migration. The filter element shall be placed in a test filter housing and then this assembly shall be mounted in the normal position (bowl at bottom) on a vibration test stand. The filter assembly shall be filled to the top of the filter element with pre-cleaned test fluid (see

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4.4.1.1), capped at both inlet and outlet ports, and soaked at 350° ± 5°F for 100 hours. The filter assembly shall then be cooled to 70° to 90°F and subjected to a sinusoidal vibration test in three mutually perpendicular axes. The rate of change of frequency shall be logarithmic.

4.6.8.1 Resonance search. Determine filter element resonant frequencies by varying the frequency of applied vibration as specified in part 1 of table V. Repeat the resonance search for each of the three axes.

TABLE V. Vibration data

Part	Vibration level (double amplitude inch or gravity)	Frequency band (Hz)
1	0.1 inch 1.0g 0.036 inch 10.0g	5 to 14 14 to 23 23 to 74 74 to 2,000
2	0.1 inch 1.0g 0.06 inch 20.0g	5 to 14 14 to 23 23 to 104 104 to 2,000

4.6.8.2 Resonance dwell. The filter assembly shall be vibrated along each axis at the four most severe resonant frequencies for 30 minutes at each resonance. Vibration levels are given in part 1 of table V and vibration times are given in part 1 of table VI. If a new resonant frequency occurs during this test, adjust the vibration frequency to maintain peak resonance.

TABLE VI. Vibration schedule

Part	Total vibration time at resonance (min)	Number of resonances
1	0 30 60 90 120	0 1 2 3 4
2	30 minutes cycling per axis	0

4.6.8.3 Cycling test. The filter assembly shall be vibrated along each axis at the vibration level and frequencies in part 2 of table V for a total of 3 hours, less the resonance dwell time for each axis in 4.6.8.2. Vibration frequencies shall be logarithmically swept from 5 to 2,000 to 5 Hz during a 20 minute interval.

4.6.8.4 Determine media migration. Remove inlet and outlet port caps and, without loss of test fluid, install the filter element in a test system (a typical test system is shown in figure 11). Close the test system, pressurize the pre-cleaned test fluid to obtain rated flow, pass one gallon of the

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test fluid through the filter element, and then pass the test fluid through a previously cleaned 0.8 micron absolute filter membrane (see 6.6). The filter membrane shall be examined under a microscope and there shall be no filter element media migration. There shall be no sign of structural failure indicated through a visual examination and the bubble point test in 4.6.2.1.

4.6.9 Filter element pressure relief valve.

4.6.9.1 M-83870-6 pressure relief valve (see figure 5). The filter element media shall be blocked, the filter element installed in a test filter housing, and placed in a test system filled with test fluid pre-cleaned in accordance with 4.4.1.1. Test fluid pressure shall be applied to the inlet port of the test filter housing and shall be increased until the cracking (opening) pressure of the pressure relief valve is reached. The cracking pressure shall occur at 100 ± 10 psi. The test fluid pressure at the inlet port of the test filter housing shall be increased until the test fluid flow rate through the pressure relief valve is 11 gal/min. The differential pressure between the inlet and outlet ports of the test filter housing at this flow rate shall be 160 psi maximum. The test fluid pressure at the inlet port of the test filter housing shall be reduced until the pressure relief valve reseats (closes). The pressure relief valve reseating pressure shall be 65 psi minimum.

4.6.9.2 M-83870-7 pressure relief valve (see figure 6). The filter element media shall be blocked, the filter element installed in a test filter housing, and placed in a test system filled with test fluid pre-cleaned in accordance with 4.4.1.1. Test fluid pressure shall be applied to the inlet port of the test filter housing and shall be increased until the cracking (opening) pressure of the pressure relief valve is reached. The cracking pressure shall occur at 50 ± 10 psi. The test fluid pressure at the inlet port of the test filter housing shall be increased until the test fluid flow rate through the pressure relief valve is 3 gal/min. The differential pressure between the inlet and outlet ports of the test filter housing at this flow rate shall be 75 psi maximum. The test fluid pressure at the inlet port of the test filter housing shall be reduced until the pressure relief valve reseats (closes). The pressure relief valve reseating pressure shall be 38 psi minimum.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements.

Packaging requirements are maintained by the Inventory Control Point packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department or Defense Agency automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Test fluid residue. Sample filter elements used for conformance tests shall be free of all contaminants and test fluid prior to packaging.

6. NOTES

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(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The filter elements covered by this specification are intended for use as follows:

		Figure
a. M83870-1	F-100 CSD Charge	1
b. M83870-2	F-100 CSD Scavenge	1
c. M83870-3	KC-135A CSD Charge, B-52F CSD Charge	2
d. M83870-4	B-52G CSD Charge, B-52G CSD Scavenge, C-5 CSD Charge, F-106 CSD Charge	3
e. M83870-5	A-7 CSD Charge, C-141 CSD Charge, F-111 CSD Charge	4
f. M83870-6	Alternate A-7 CSD Charge, C-141 CSD Charge, F-111 CSD Charge	5
g. M83870-7	F-4 CSD Charge	6
h. M83870-8	EC/R-135 CSD Charge, B-52H CSD Charge	7
i. M83870-9	C-135B CSD Charge	8
j. M83870-10	C-141 G.E. CSD Charge	9

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Item identification (see 3.7, items e and f).
c. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
d. Part number (see 1.2).
e. Micron rating (see table II).
f. If the conformance inspection sampling inspection level is other than S-1 (i.e., normal, tightened, or reduced) (see 4.3.2).
g. Packaging requirements (see 5).
h. If qualification data is required (see 4.2).
i. If multi-pass efficiency testing is required (see 4.6.4.5)

6.3 Qualification. The attention of the contractors is called to the requirements with respect to products requiring qualification. Awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the Qualified Products List (QPL No. 83870) whether or not such products have actually been listed by that date. In order that the manufacturers may be eligible to be awarded contracts or purchase orders for the products covered by this specification, they are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification. Information pertaining to qualification of products may be obtained from Oklahoma City Air Logistics Center/TICLA, 3001 Staff Drive, Suite 1AE1-101A, Tinker AFB, Oklahoma 73145-3036.

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6.3.1 Efficiency testing. Future changes to this specification will incorporate multi-pass efficiency testing, in accordance with ISO 4572 which yields a Beta ratio, and ultimately a filter element efficiency rating at various particle sizes. Contractors seeking qualification of their filter elements are encouraged to incorporate multi-pass efficiency testing along with the existing efficiency test. Multi-pass efficiency testing is also encouraged when QPL items require re-qualification.

6.4 Filter. Type MIL-F-27656 or equivalent.

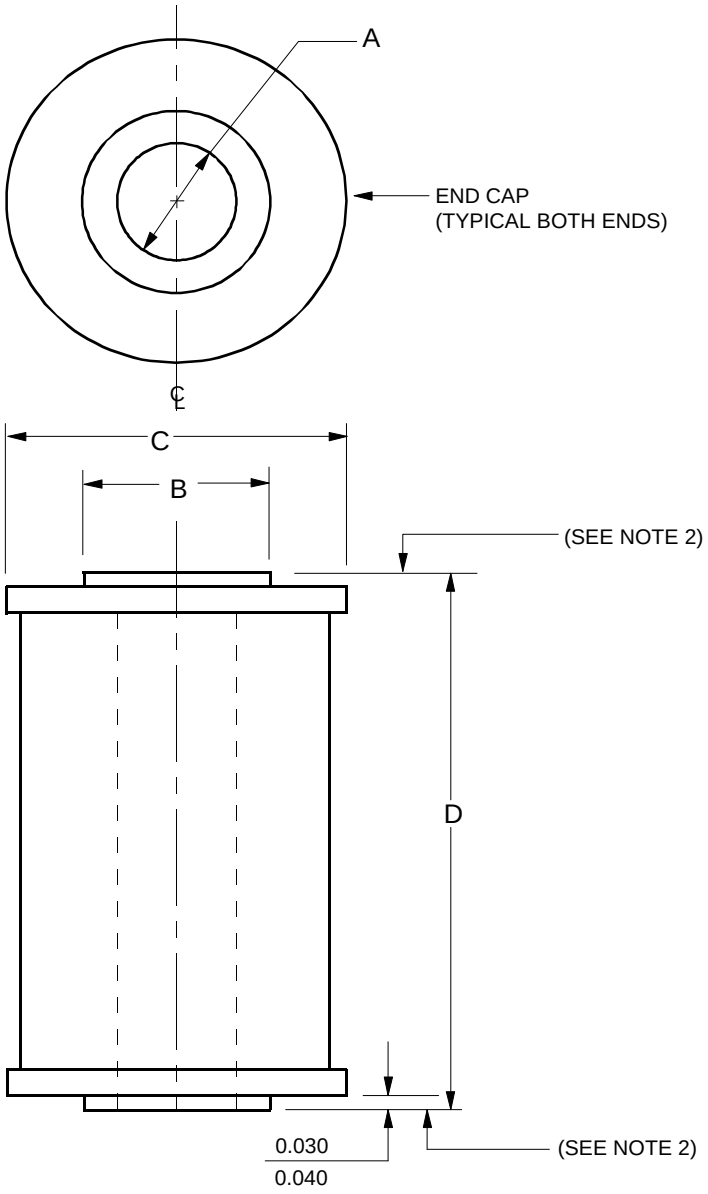
6.5 Filter. Type HA Millipore or equivalent.

6.6 Filter. Type AA Millipore or equivalent.

6.7 Subject term (key word) listing.
Beta ratio
Bubble point
Filter element efficiency
Flow fatigue
Immersion
Lubricating oil
20 Micron
Multi-pass efficiency

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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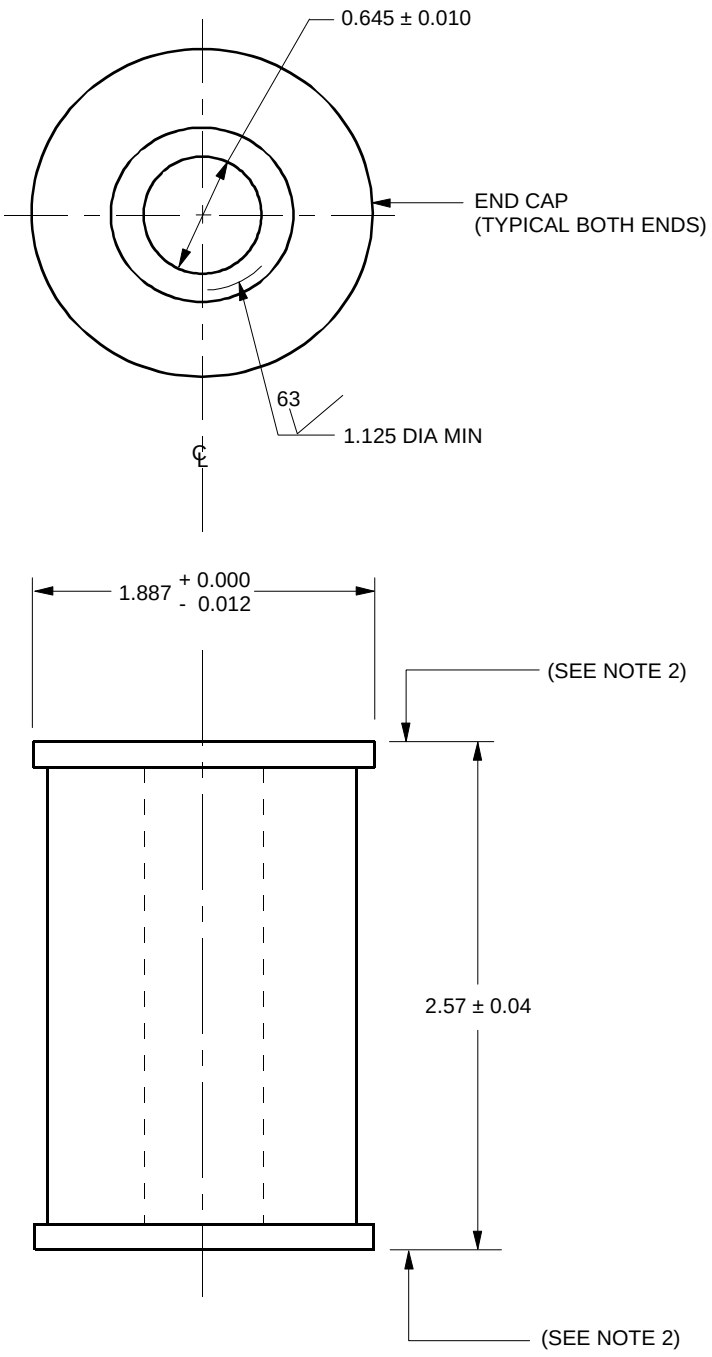


Part Number	A ± 0.010	B ± 0.020	C ± 0.006	D ± 0.020
M83870-1	0.645	1.120	1.881	1.970
M83870-2	0.645	1.120	1.881	3.980

NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. SURFACES SHALL BE PARALLEL WITHIN 0.010 TOTAL INDICATOR READING.
3. THIS DRAWING IS NOT TO SCALE.

FIGURE 1. M83870-1, -2 filter element dimensions

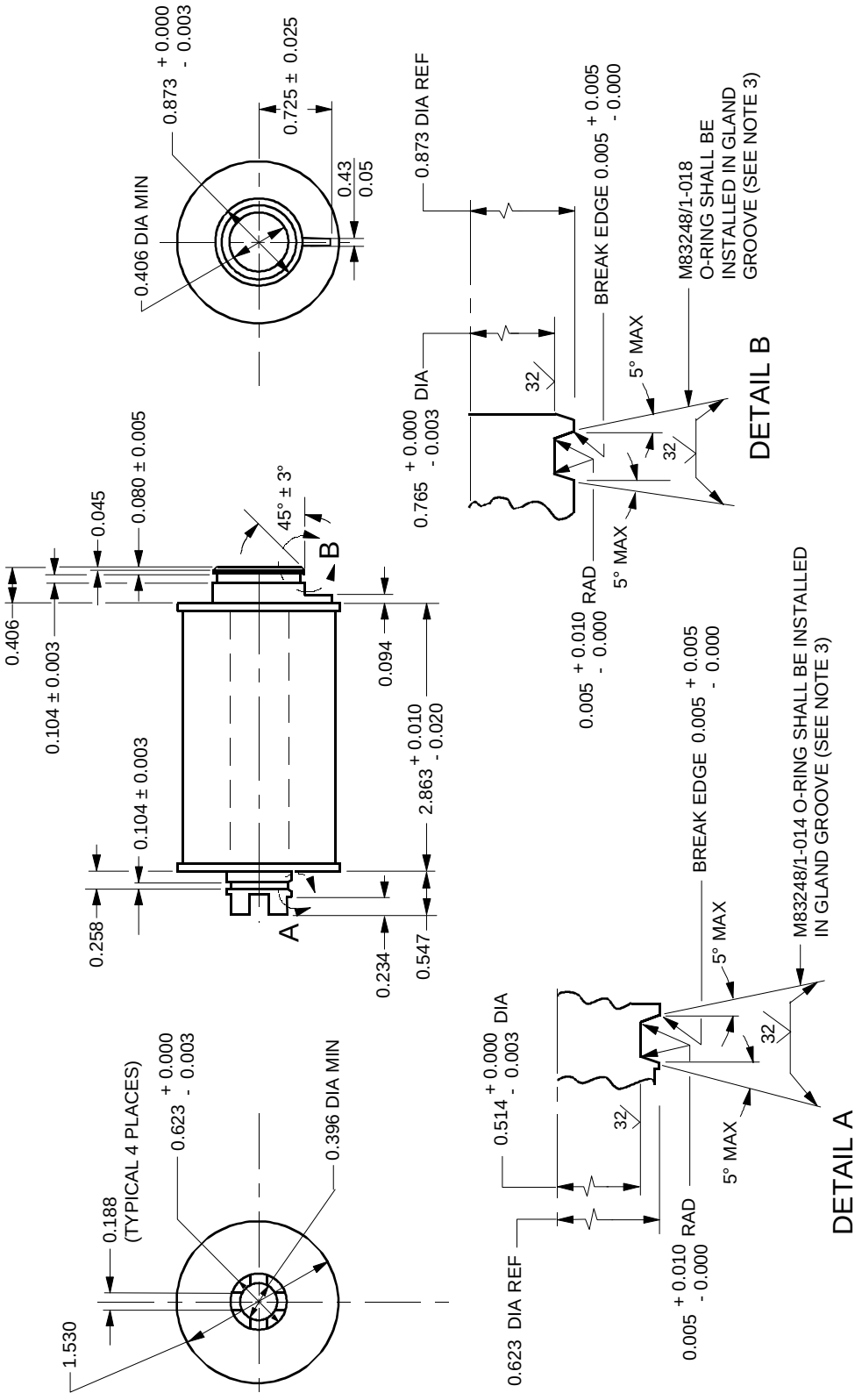
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NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. SURFACES SHALL BE PARALLEL WITHIN 0.010 TOTAL INDICATOR READING.
3. THIS DRAWING IS NOT TO SCALE.

FIGURE 2. M83870-3 filter element dimensions

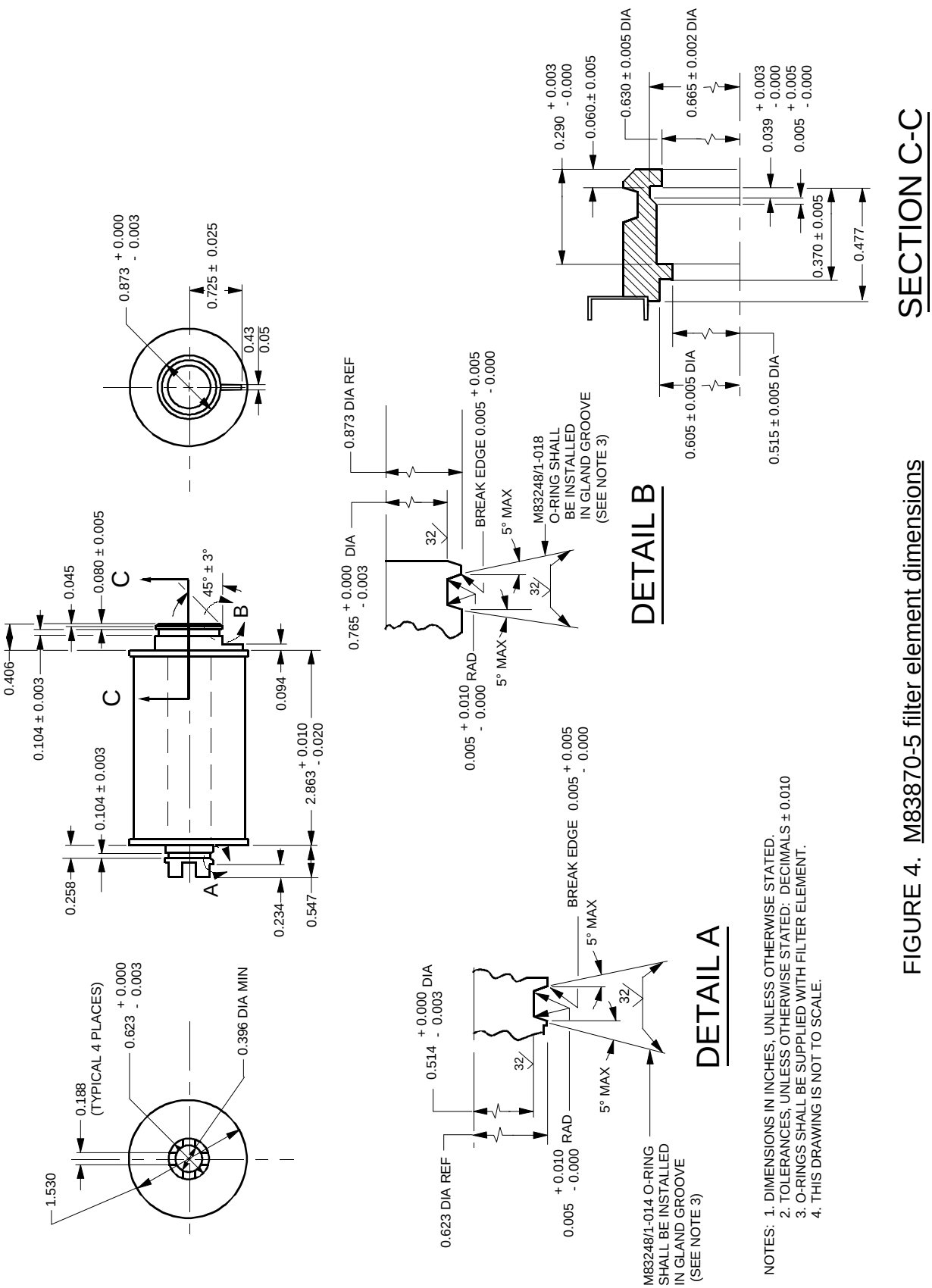
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NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCES, UNLESS OTHERWISE STATED: DECIMALS ± 0.010
3. O-RINGS SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. THIS DRAWING IS NOT TO SCALE.

FIGURE 3. M83870-4 filter element dimensions

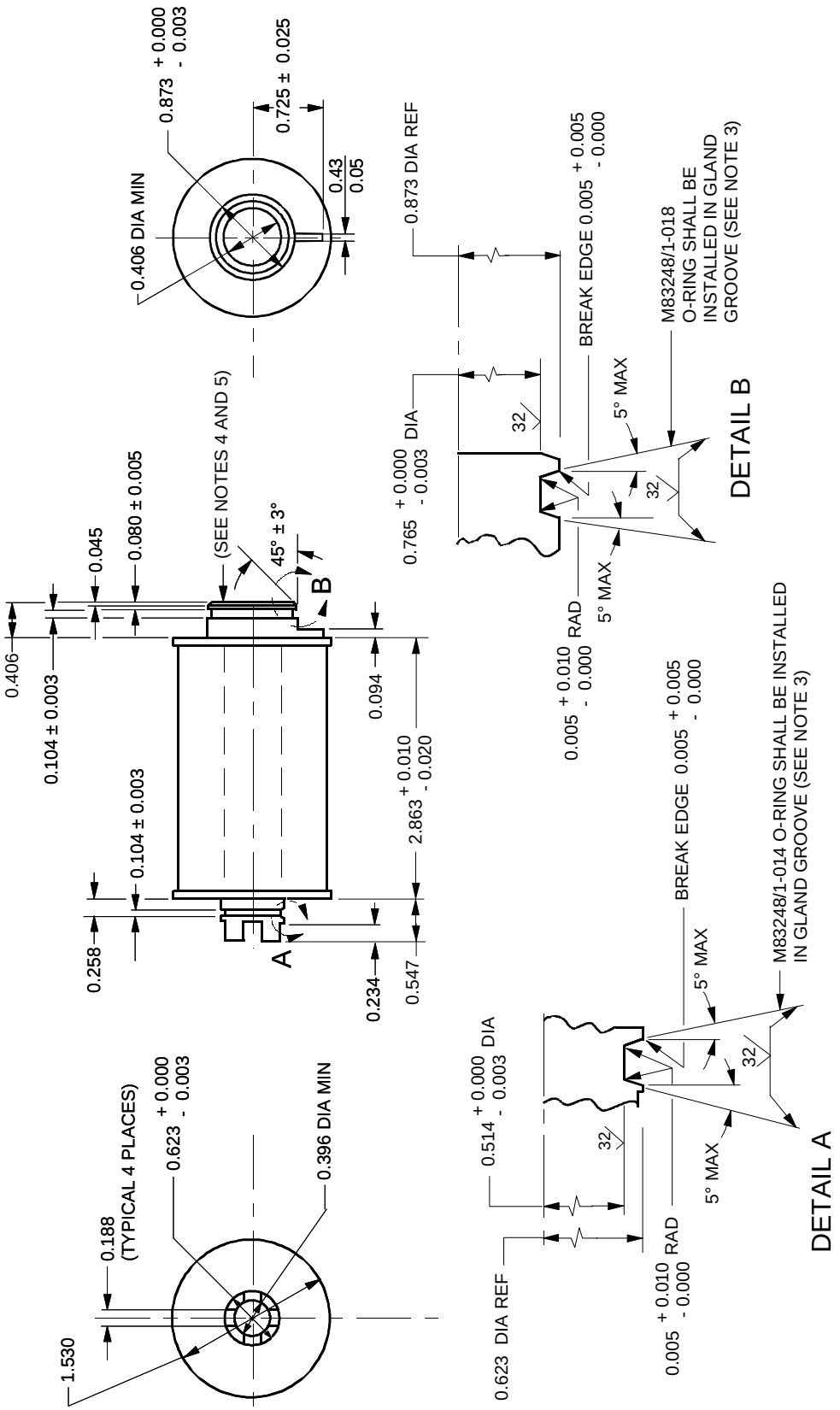
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NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCES, UNLESS OTHERWISE STATED: DECIMALS ± 0.010
3. O-RINGS SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. THIS DRAWING IS NOT TO SCALE.

FIGURE 4. M83870-5 filter element dimensions

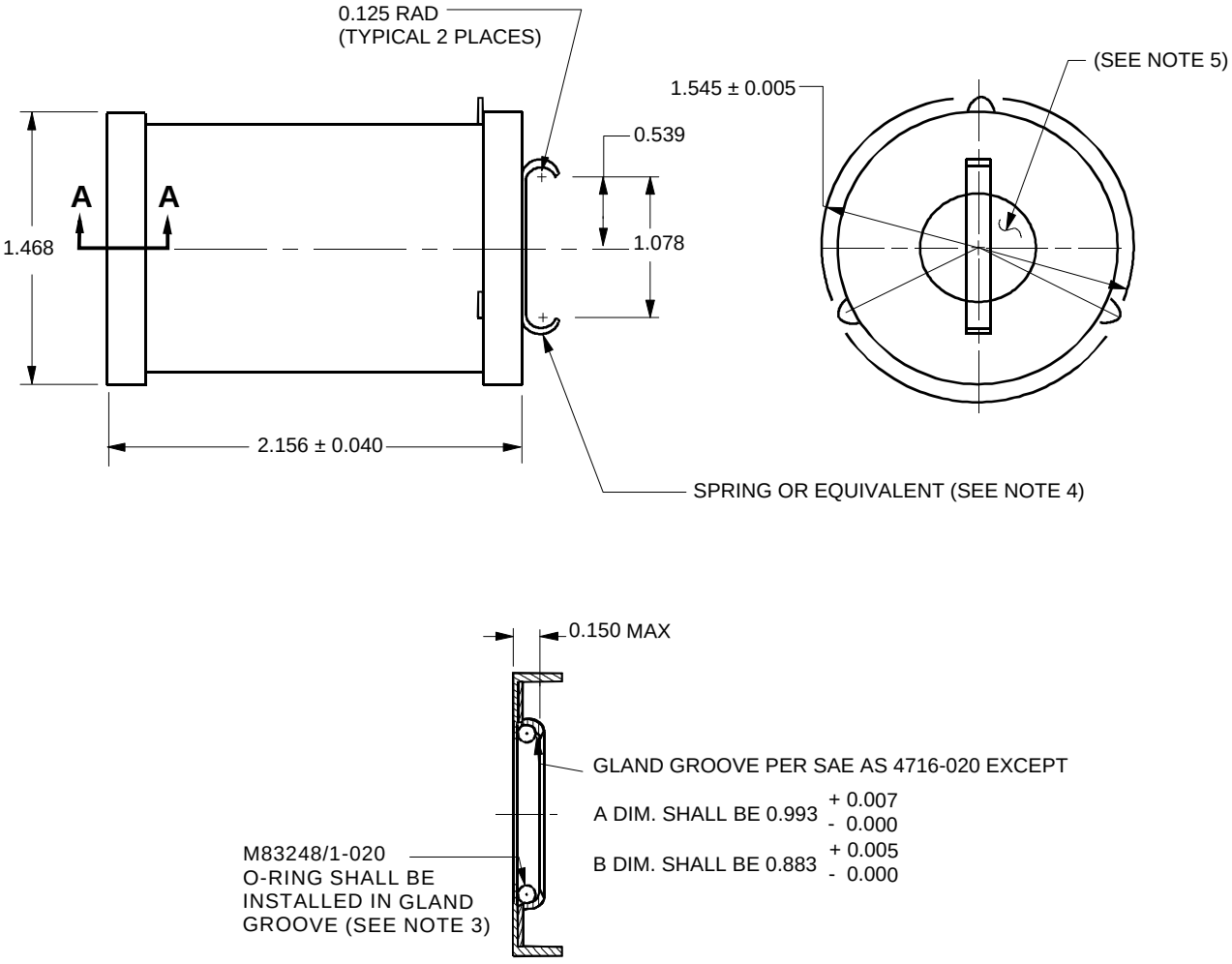
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NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCES, UNLESS OTHERWISE STATED: DECIMALS ± 0.010.
3. O-RINGS SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. RELIEF VALVE THIS END.
5. RELIEF VALVE HOUSING SHALL HAVE A DIAMETER OF 0.780 MAX AND SHALL EXTEND 0.426 MAX FROM END CAP.
6. THIS DRAWING IS NOT TO SCALE.

FIGURE 5. M83870-6 filter element dimensions

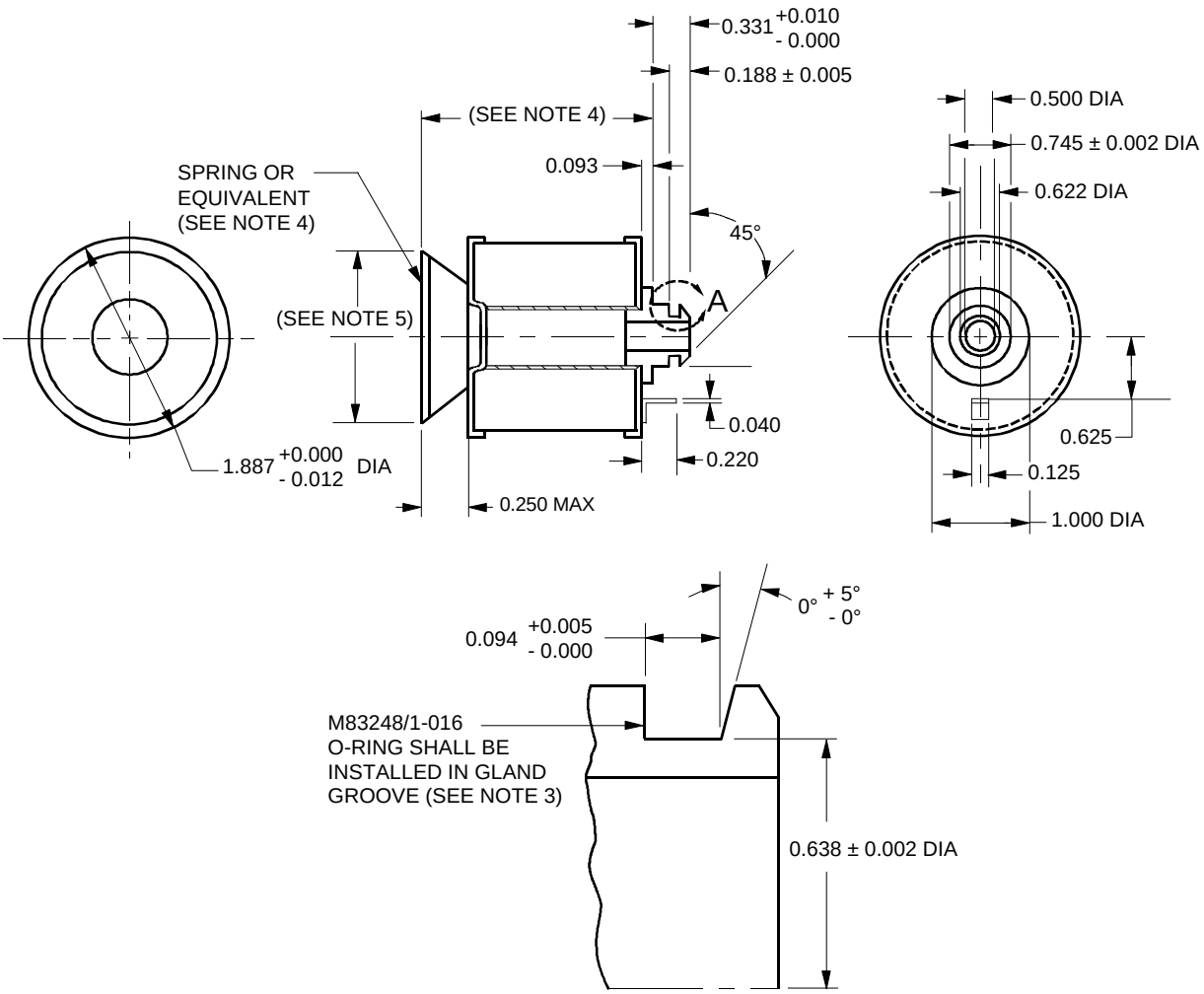
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NOTES:
1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCES, UNLESS OTHERWISE STATED: DECIMALS ± 0.010.
3. O-RING SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. SPRING CHARACTERISTICS MUST BE SUCH AS TO PROVIDE AXIAL LOADING OF 5 TO 15 POUNDS WHEN COMPRESSED 0.09 INCH.
5. RELIEF VALVE THIS END.
6. THIS DRAWING IS NOT TO SCALE.

FIGURE 6. M83870-7 filter element dimensions

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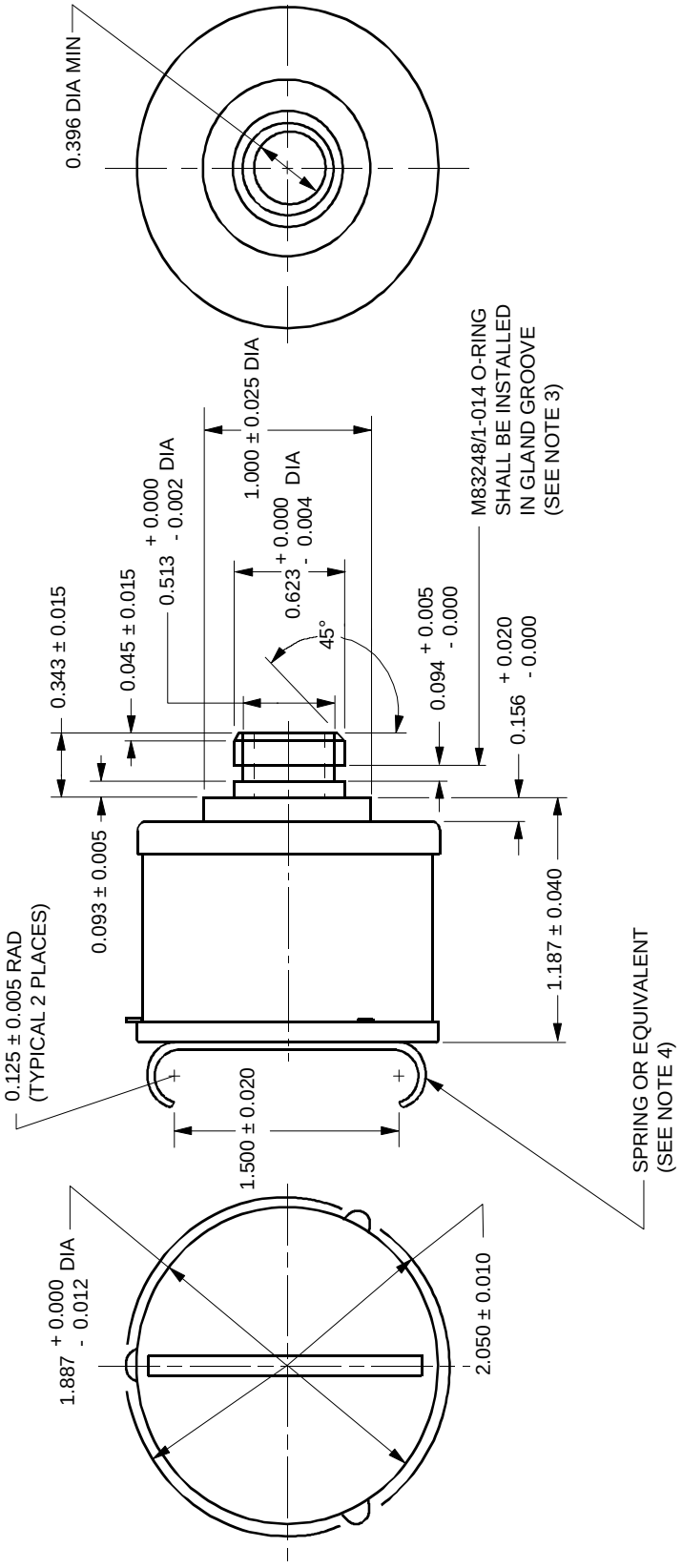


DETAIL A

NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCES: DECIMALS ± 0.010
ANGLES ± 3°
3. O-RING SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. ELEMENT LENGTH AND SPRING CHARACTERISTICS MUST BE SUCH AS TO PROVIDE AXIAL LOADING OF 10 TO 50 POUNDS WHEN COMPRESSED TO 1.670 ± 0.010 INCH.
5. MAXIMUM DIA WHEN COMPRESSED TO 1.670 ± 0.010 INCH IS 1.75 INCH. MINIMUM DIA WHICH WILL CONTACT FILTER BOWL IS 0.812 INCH.
6. THIS DRAWING IS NOT TO SCALE.

FIGURE 7. M83870-8 filter element dimensions

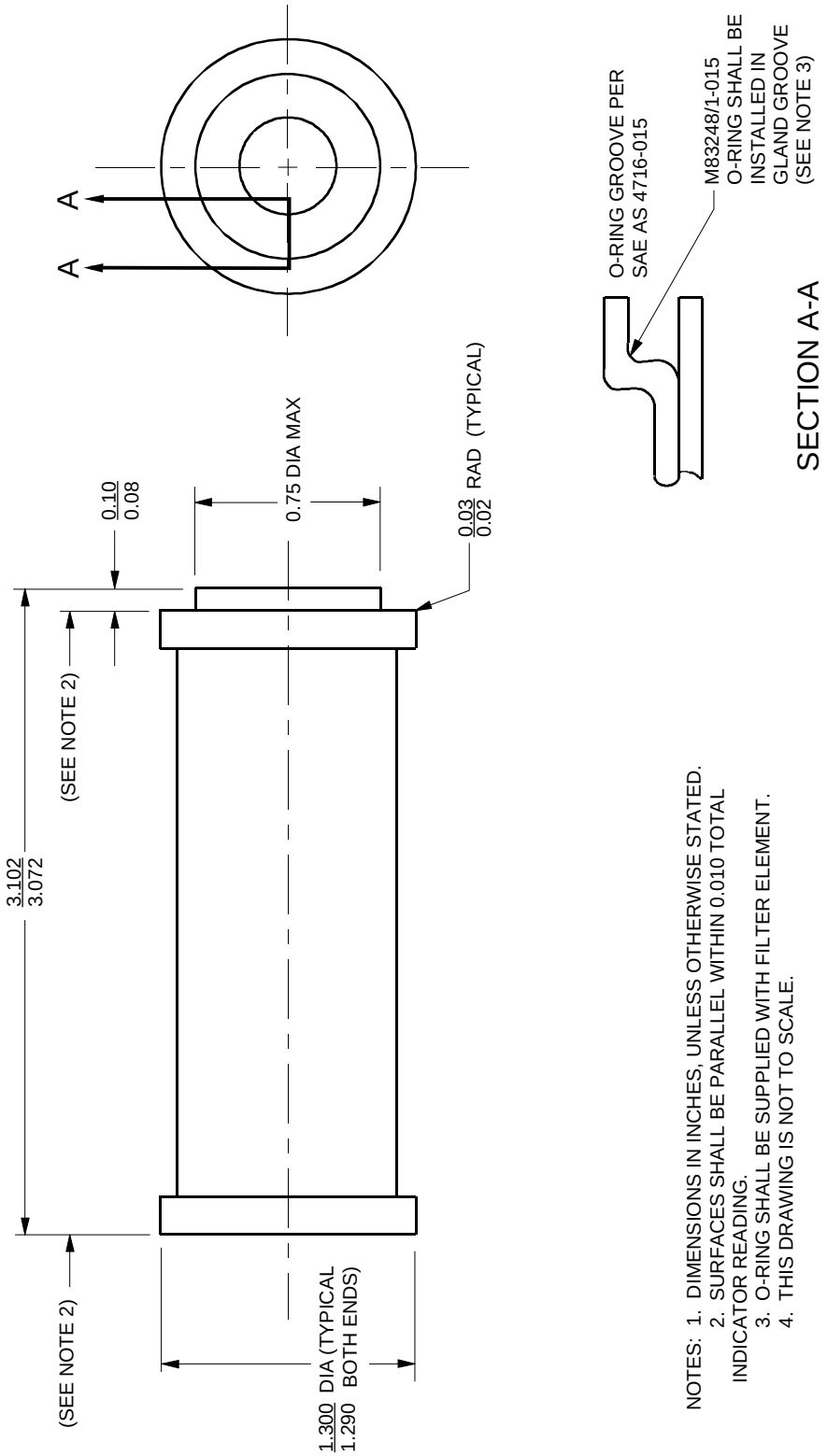
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NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. TOLERANCE: ANGLE ± 3°.
3. O-RING SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. SPRING CHARACTERISTICS MUST BE SUCH AS TO PROVIDE AXIAL LOADING OF 5 TO 15 POUNDS WHEN COMPRESSED 0.09 INCH.

FIGURE 8. M83870-9 filter element dimensions

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NOTES: 1. DIMENSIONS IN INCHES, UNLESS OTHERWISE STATED.
2. SURFACES SHALL BE PARALLEL WITHIN 0.010 TOTAL INDICATOR READING.
3. O-RING SHALL BE SUPPLIED WITH FILTER ELEMENT.
4. THIS DRAWING IS NOT TO SCALE.

FIGURE 9. M83870-10 filter element dimensions

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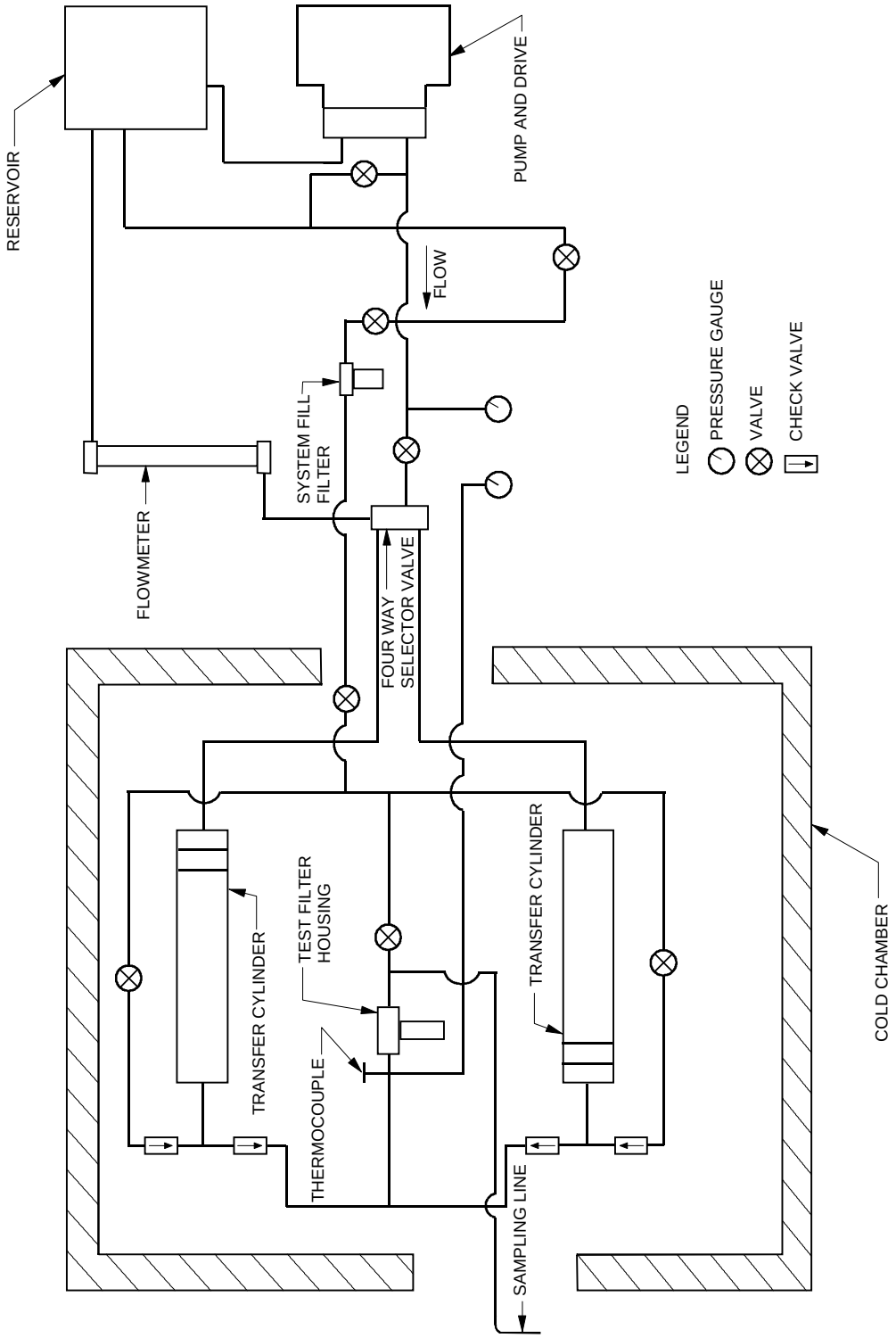


FIGURE 10. Typical setup for cold start test

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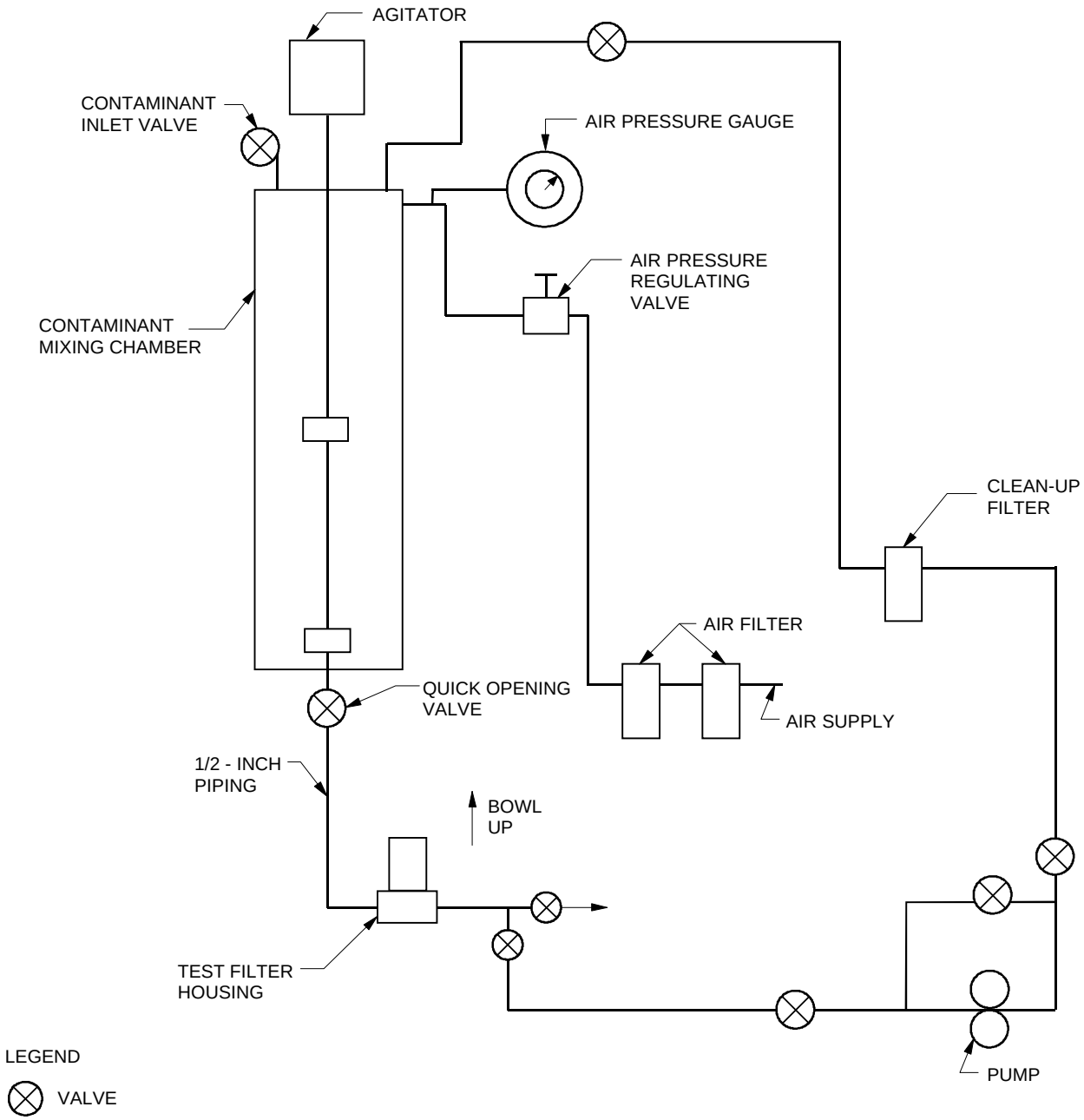
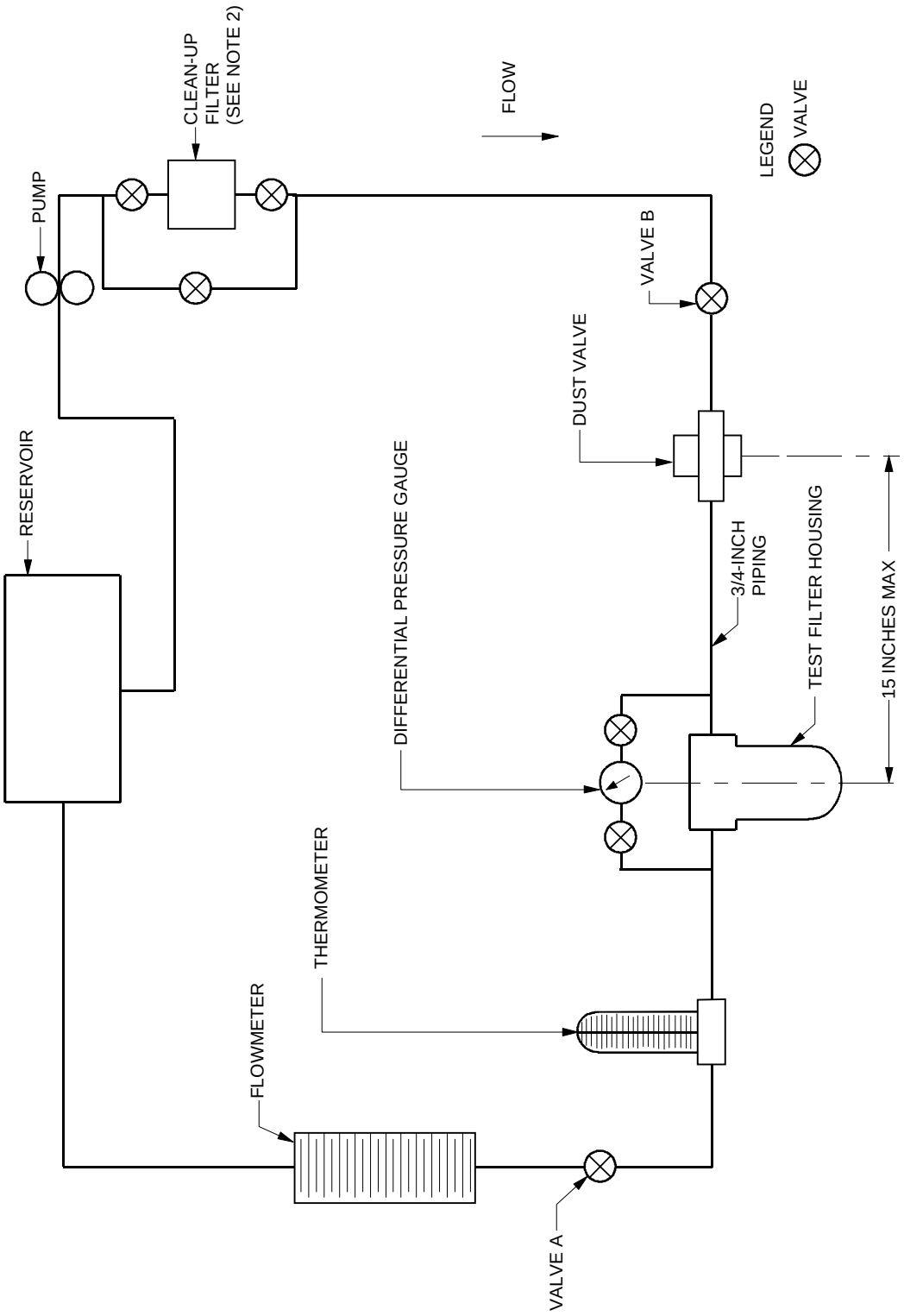


FIGURE 11. Typical setup for efficiency test

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NOTES: 1. TOTAL SYSTEM VOLUME SHALL NOT EXCEED 15 GALLONS.
2. CLEAN-UP FILTER SHALL BE BYPASSED DURING PRESSURE BUILDUP AND COLLAPSE TEST.

FIGURE 12. Typical setup for pressure buildup and collapse test

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	MIL-PRF-83870B(USAF)	**970627**

3. DOCUMENT TITLE

FILTER ELEMENT, FLUID PRESSURE, TRANSMISSION, AIRCRAFT, DISPOSABLE, 20 MICRON ABSOLUTE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. Organization	
c. ADDRESS *(Include zip code)*	**d. TELEPHONE** *(Include Area Code)* (1) Commercial (2) Autovon *(if applicable)*	**7. DATE SUBMITTED** **(YYMMDD)**

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE** *(Include Area Code)*	
Harlena Edwards	(1) Commercial (405) 736-5960	(2) Autovon 336-5960
b.* ADDRESS *(Include Zip Code) **OC-ALC/TICLA** **3001 Staff Drive Ste 1AE1-101A** **Tinker AFB, OK 73145-3029**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3446** **Telephone (703) 756-2340 Autovon 289-2340**	

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